

Task 3: Cuisine Classification

Objective

The objective of this task is to develop a machine learning model capable of classifying restaurants based on their cuisine type using restaurant-related features available in the dataset.

Dataset Description

The dataset contains restaurant information such as:

- Restaurant Name
- City
- Locality
- Average Cost for Two
- Price Range
- Online Delivery Status
- Table Booking Status
- Votes
- Aggregate Rating
- Cuisine Type

The target variable for classification is the cuisine category of each restaurant.

Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the restaurant dataset.
2. Removed records with missing cuisine values.
3. Selected the primary cuisine for restaurants offering multiple cuisines.
4. Encoded categorical variables using Label Encoding.
5. Split the dataset into training and testing sets using an 80:20 ratio.

Model Selection

A Random Forest Classifier was used for cuisine classification. Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Model Training

The model was trained using the training dataset and evaluated using the testing dataset.

Performance Evaluation

Accuracy

Accuracy = **40.28%**

Classification Metrics

The model performance was evaluated using:

- Precision
- Recall
- F1-Score
- Accuracy

Weighted Average Results:

- Precision: **0.37**
- Recall: **0.40**
- F1-Score: **0.34**

Analysis

The model achieved an overall accuracy of 40.28%. The dataset contains a large number of cuisine categories, with some cuisines having very few restaurant samples. This class imbalance affected the model's ability to correctly predict less frequent cuisines. Cuisine classes with a higher number of samples achieved better prediction performance, while classes with very limited samples showed lower precision and recall values. The warnings generated during evaluation occurred because certain cuisine categories had insufficient examples in either the training or testing datasets.

Challenges

- High number of cuisine categories.
- Imbalanced distribution of cuisine classes.
- Some cuisine categories contained very few records.
- Difficulty in accurately classifying rare cuisine types.

Conclusion

A cuisine classification model was successfully developed using the Random Forest Classification algorithm. The model achieved an accuracy of 40.28% on the test dataset. Although the performance was affected by class imbalance and the large number of cuisine categories, the model demonstrated the ability to classify common cuisine types effectively. Future improvements can be achieved through data balancing techniques, feature engineering, and hyperparameter optimization.